

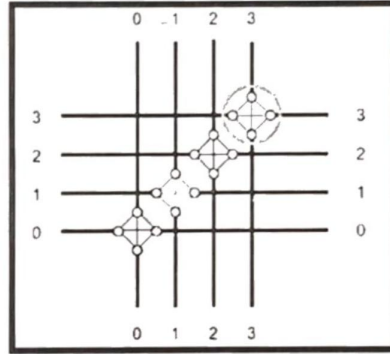
Q1. The following functions

[4M]

- $f_1(w, x, y, z) = \sum m(0, 2, 5, 7, 8, 10, 12, 13)$
- $f_2(w, x, y, z) = \sum m(0, 2, 6, 8, 9, 14, 15)$
- $f_3(w, x, y, z) = \sum m(0, 8, 14, 15)$
- $f_4(w, x, y, z) = \sum m(0, 1, 2, 4, 5, 8, 9, 10)$

Implement four logic functions using a minimum-size PLA (Draw the diagram). Mention the size of the PLA obtained?

Q2. Draw a possible implementation (detailed structure/circuit) of a bidirectional switch (circled one) in an FPGA. [3M]



Q3. Consider the following logic function 'F' of four variables in its initial Sum-of-Products (SOP) form:

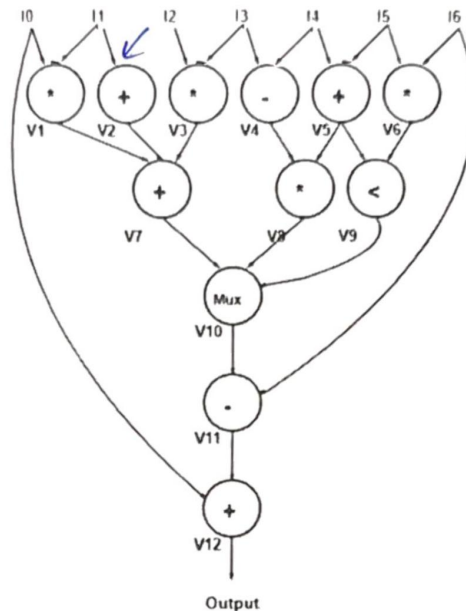
$$F(w, x, y, z) = w\bar{x}yz + w\bar{x}y\bar{z} + wx\bar{y}z + \bar{w}xz + \bar{w}\bar{x}z + wxy\bar{z}$$

What is the objective of the node manipulation technique? Mention any two node manipulation techniques, and apply the same to the function provided above (the techniques should be implemented independently, not one after the other). Which of the two node manipulation techniques applied above is better in achieving the objectives of the node manipulation technique? Explain. [4M]

Q4. Explain the key differences between the Karnaugh and Quine-McCluskey placement methods. [2M]

Q.5. A data flow graph is given below.

[5M+3M+2M+2M]



Operation details are given below

- Mux (Multiplexer): 10 clock cycles, 12 LUTs
- < (Comparison): 15 clock cycles, 18 LUTs
- + (Addition): 20 clock cycles, 25 LUTs
- - (Subtraction): 20 clock cycles, 25 LUTs
- * (Multiplication): 35 clock cycles, 100 LUTs

FPGA size: 150 LUTs

(a) Apply the list scheduling-based temporal partitioning algorithm for the sequencing graph for the above data-flow graph. The priority is assigned based on the number of successors. While performing the scheduling, ensure that the partitioning is done to minimize the total computation time of DFG (consider this while partitioning nodes of same priority). Clearly show which nodes are mapped to which partition, and calculate the LUT utilization for each partition.

(b) Calculate the wasted resources for the partitioning obtained in (a). Show the steps in the calculation

(c) Calculate the quality of the partitioning.

(d) Do you think partial reconfiguration is useful for the logic implementation given ^{above} ~~below~~? Give Justification for your answer.